Survey of Information Spread Models

Megan Mattingly

Transylvania University

Project Motivation

- Research project started as an in-class final project in Mathematical Modeling
- Created a Markov Chain matrix model and a differential equation model for information spread
- These models were extended and a third model was created

Background

- Information spread on online networks can be modeled in many different ways
- The forest-fire algorithm is based off of the modified forest-fire model (Kumar 2020), which relates information spread online to a wildfire spreading through a forest
- The algorithm was coded using R
- Users that are in the ”Tree” state represent those that are on the network, but do not know the information
- The ”Fire” state represents users that have learned about the information and are actively spreading it to others

Outline

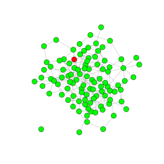
Outline

Outline

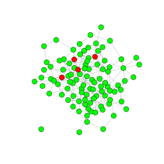
Model Formulation and Assumptions

- The algorithm creates a network graph that randomly connects nodes together with a set number of edges
- When a node is informed, they can only spread the information to nodes that they are directly connected to
- We randomly select one node to start as Fire
- The algorithm ran 100 iterations of the simulation for each version of the model

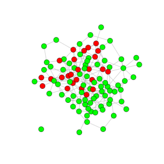
Basic Model Simulations



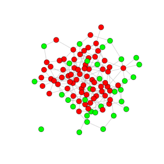
(a) Time Step 1



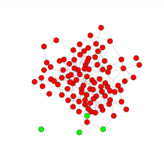
(b) Time Step 2



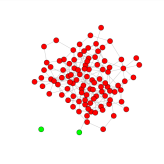
(c) Time Step 3



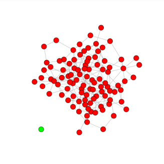
(d) Time Step 4



(e) Time Step 5



(f) Time Step 6

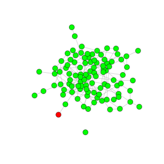


(g) Time Step 7

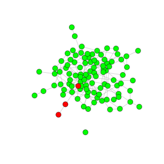
Node Probability Model

- Probability is introduced into the algorithm
- Rather than immediately spreading the information to all its neighbors, the node has the option to either spread or not
- A random probability value is assigned to each node at each time step and this is compared to a set cutoff value
- This creates more variation in the model

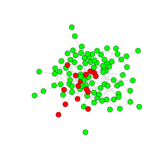
Node Probability Model Simulations



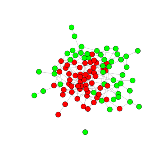
(a) Time Step 1



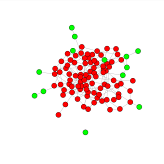
(b) Time Step 3



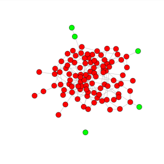
(c) Time Step 5



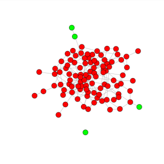
(d) Time Step 7



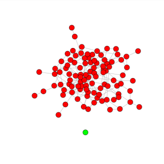
(e) Time Step 9



(f) Time Step 11



(g) Time Step 13

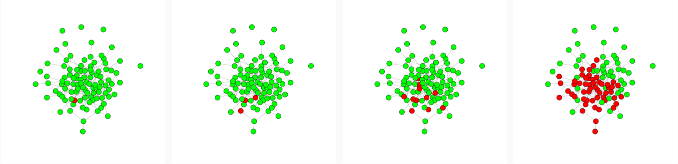


(h) Time Step 16

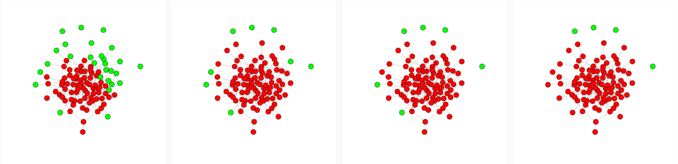
Neighbor Probability Model

- Probability is used again, but in a different way
- The node has the option to spread information or not to each individual neighbor, rather than spreading to all or none
- All neighbors of a node that is in the Fire state are randomly assigned a probability value, which is compared to a set cutoff value
- This also creates more variation in the model

Neighbor Probability Model Simulations



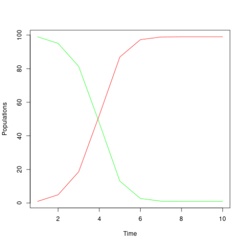
(a) Time Step 1 (b) Time Step 2 (c) Time Step 3 (d) Time Step 5



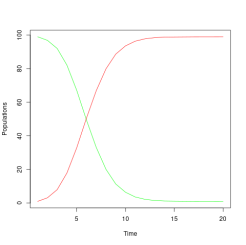
(e) Time Step 6 (f) Time Step 8 (g) Time Step 9 (h) Time Step 12

Analysis

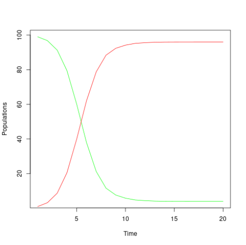
- We ran 100 stochastic simulations for each version of the model and took the average results of each



(a) Basic



(b) Node Probability

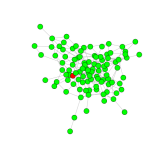


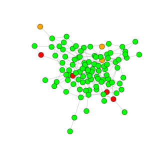
(c) Neighbor Probability

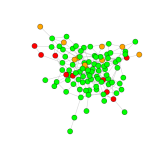
Burnt Model Extension

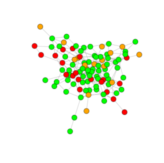
- Now, the model will be extended to include a third compartment
- ”Burnt” refers to a user that learns about the information, but chooses not to spread it any further
- Whether a node becomes Fire or Burnt is based off of a randomly assigned probability value
- This means that the information may or may not reach the whole population in this scenario

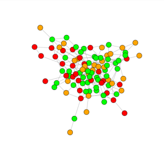
Burnt Model Simulations

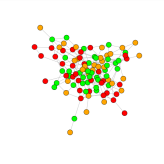

(a) Time Step 1

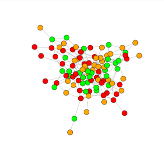

(b) Time Step 2


(c) Time Step 3


(d) Time Step 4

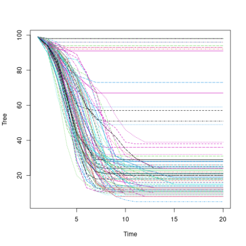

(e) Time Step 5


(f) Time Step 6

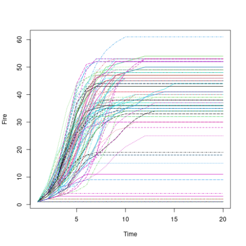

(g) Time Step 7

Analysis

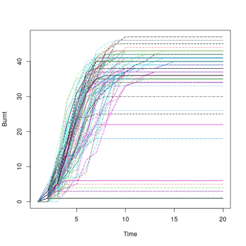
- Stochastic simulations for the burnt extension model are shown below
- More variation in results for the final class proportions



(a) Tree



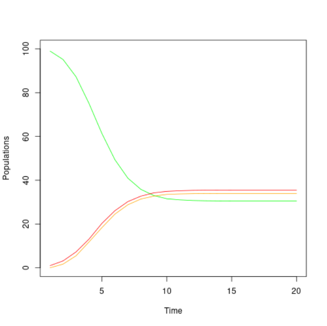
(b) Fire



(c) Burnt

Analysis

- We look at the average results of all the simulations to see the overall trend



Limitations and Conclusion

- Although different versions of the algorithm incorporate different assumptions, there are still things that it cannot account for
- This algorithm doesn't allow users to move between states
- Other models include a reinfection rate to account for this
- This algorithm is able to show how small changes in user behavior affects information spread